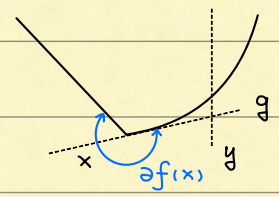
Subgradient 次梯度

Subdifferential $\partial f(x) := \{ g \mid f(y) \geq f(x) + \langle y-x, g \rangle \}$

Subgradient $g \in \partial f(x)$



e.g. $f(x) = \|x\|$,

$\partial f(0) = \mathrm{Ball}_{\|\cdot\|_*} = \{ v \in E^* \mid \|v\|_* \leq 1 \}$

Pf: Holder ineq $\langle v, y \rangle \leq \|v\|_* \cdot \|y\| \leq \|y\|$

Dual norm $\|v\|_* = \sup \frac{\langle x, v \rangle}{\|x\|} = \sup_{\|x\| \leq 1} \langle x, v \rangle$

$\partial f(x) = \{ v \in E^* \mid \langle x, v \rangle = \|x\|, \ \|v\|_* \leq 1 \}$

Pf: $\Leftarrow$ 要证 $\|x\| + \langle v, y \rangle - \langle v, x \rangle \leq \|y\|, \ \forall y$

由 $\langle x, v \rangle = \|x\|$，只需证 $\langle v, y \rangle \leq \|y\|, \ \forall y$

Holder ineq $\langle v, y \rangle \leq \|v\|_* \cdot \|y\| \leq \|y\|$

$\Rightarrow$ 比较难证，需要 Conjugate of norm

Conjugate $\|v\|^* = \sup \langle x, v \rangle - \|x\|$

$= \delta_{B_{\|\cdot\|_*}[0,1]}(v) = \begin{cases} 0, & \|v\|_* \leq 1 \\ \infty, & \|v\|_* > 1 \end{cases}$

$f(y) \geq f(x) + \langle v, y-x \rangle, \ \forall y$

$\|y\| \geq \|x\| + \langle v, y \rangle - \langle v, x \rangle, \ \forall y$

$\langle v, x \rangle - \|x\| \geq \langle v, y \rangle - \|y\|$ $\sup_y$ 即为 $\|v\|^*$

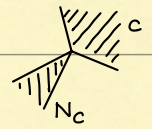
$\geq \begin{cases} 0, & \|v\|_* \leq 1 \\ \infty, & \|v\|_* > 1 \end{cases}$

由于 $\langle v, x \rangle - \|x\|$ 必有限，故必有 $\langle v, x \rangle - \|x\| \geq 0$ 和 $\|v\|_* \leq 1$

又 $0 \leq \langle v, x \rangle - \|x\| \leq \|v\|_* \|x\| - \|x\| \leq 0$

故 $\langle v, x \rangle - \|x\| = 0$

$f(x) = I_C(x)$, $\partial f(x) = N_C(x)$



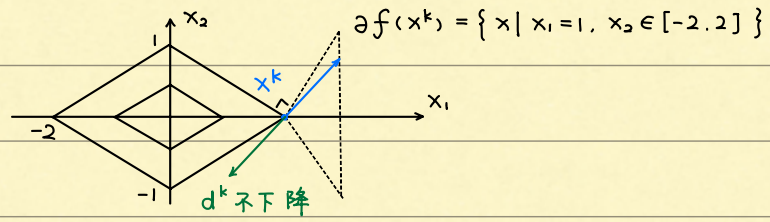
Normal Cone

Fermat Optimal Condition：若 f 凸，$0 \in \partial f(x^*)$

Pf：convex，$f(x) \geq f(x^*) + \langle x - x^*, 0 \rangle$

Subgradient Descent

$$x^{k+1} = x^k - \alpha_k g^k, \quad g^k \in \partial f(x^k)$$

次梯度法不一定下降

e.g. $f(x) = |x_1| + 2|x_2|$